

A 10,240-Byte Generator, Scored After It Is Selected

The DOPE Project

Abstract—DOPE’s measured generator on this panel is a compact residual network: at most 12 inputs, one hidden layer of width 16 with $\tanh$ units, and a refit linear readout. AdamW trains the hidden basis for 2,048 steps at learning rate 0.002. The shipped network has no embedding layer. Each method is selected on its own objective. The shared measurements are null-normalized retention at four times the fit size, and the bytes charged to the artifact. On 97 informative PMLB regression lineages, the displayed DOPE profile has median CatBoost retention 0.939876 against 0.655681 for a native-selected Gaussian copula, 0.585986 for a study Chow–Liu tree, and -0.011971 for study independent marginals. Ninety-eight of 100 DOPE artifacts are within the 10,240-byte cap; 62, 31, and 94 of the three comparators are. A separate 21-lineage block compares DOPE with native-selected CTGAN and TVAE. Every successful CTGAN and TVAE artifact in that ledger exceeds the cap. Master Fitness v2 stays null: the six components and the hard-gate map are not all present. Official tests remain sealed. No paired superiority claim is made. Appendix A lists every one of the 100 lineages. A separate 8,192-step budget on the same partitions is reported in Appendix B and is not adopted as the displayed model. BeyondArena is a separate 142-family corpus in the dataset bucket. It is not pooled with these 100 lineages. Five of twelve locked families were fit. Seven did not prepare, and none of the five has a retention score.

Index Terms—Synthetic tabular data, compact generator, retention, artifact bytes.

I. INTRODUCTION

A generator paper usually reports the metric that was also used to pick the configuration. CTGAN and TVAE were introduced with a conditional generator and a variational autoencoder, and later libraries score them with downstream predictive efficacy [1]. The Synthetic Data Vault fits Gaussian-copula models inside that family [2]. Adversarial random forests select a density through their forest density estimator [3]. Diffusion and CART synthesizers bring other objectives [4], [5]. Those native numbers live on different scales.

The comparison here separates selection from the common measurement. DOPE, the Gaussian copula, CTGAN, and TVAE keep the objectives published for those methods. Chow–Liu and independent marginals are study implementations of the published ideas, not the authors’ packages [6]. After selection, every retained artifact is measured by the same retention formula and by the bytes required to sample from it. The pre-registered common scalar, Master Fitness v2 (MFS-v2), is defined in Section III. It is emitted only when every hard gate passes. On this panel the scalar is null for every method.

The decision the measurements can support is narrower than “which generator is best.” It is whether a network small enough to ship under a 10,240-byte cap still preserves downstream regression utility on real validation rows. Fig. 1 is that comparison. TabDDPM and synthpop CART are cited and have not been run on this panel. An unexecuted method is not a win for DOPE.

II. THE NETWORK THAT WAS TRAINED

The displayed profile is `features12_steps2048`. It is a compact neural residual, not a GAN, a diffusion model, or a transformer. The training code selects at most 12 screened inputs. A hidden layer of width 16 maps those inputs through $\tanh$. AdamW, learning rate 0.002, runs for 2,048 steps on the regression mean-squared error, with gradient clipping at 5. The hidden weights and biases from that fit are kept. The intercept, the linear skip coefficients, and the hidden readout are then refit by the same regularized solver used for the native linear candidates, on the frozen $\tanh$ features. Coefficients whose absolute value is below 10^{-5} are dropped from the skip. For a selected input vector x_S the shipped predictor is

$$h_j = \tanh(b_j + w_j^\top x_S), \quad \hat{y} = a + \beta^\top x_S + v^\top h. \quad (1)$$

Sampling adds the noise model stored in the artifact. The codec accepts hidden width 8 or 16 at the input counts used here and rejects widths 0, 17, and 32. There is no embedding table in (1). A separate dataset-action router uses a vector of dimension $856 + 24 \times (128 + 10) = 4,168$. That vector does not generate the rows in this paper.

The profile is a pre-registered reference. Compile-time search records the proxy $0.45u + 0.25d + 0.30j$ and then keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. Every such compile is marked `production_evidence_unmeasured`. No global DOPE family has been selected, so the shared retention below was not allowed to choose the profile.

Other DOPE shapes exist and are not this result. Micro-TVAEs use latent and hidden widths 4/16, 8/24, and 12/32. TinyMAT uses widths 16 and 24. Joint TVAE, masked-transformer, and TabSyn candidates use latent width 32 and hidden width 128. A direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier. The published fits did not retain a per-step loss series, so this paper does not show a training curve.

On six confirmation lineages, disjoint from the discovery set, two longer profiles were already measured at $4n$. CatBoost median retention is 0.939543 for the reference above, 0.937693 for a width-16 profile trained for 8,192 steps, and 0.949384 for a width-8 profile trained for 8,192 steps. That is a six-lineage ablation. It does not replace the reference.

III. TWO SCORES

Let L_{null} , L_{real} , and $L_{\text{synthetic}}$ be the auditor’s loss on real validation rows when that auditor is trained on a null baseline, on real fit rows, and on synthetic rows. Retention is

$$\frac{L_{\text{null}} - L_{\text{synthetic}}}{L_{\text{null}} - L_{\text{real}}}. \quad (2)$$

TABLE I
ILLUSTRATION OF (3). THESE COMPONENTS ARE NOT MEASUREMENTS.

Gate map	Scalar
All six gates passed	82.40
Byte gate failed	null

A value of 1 means the synthetic-trained auditor matched the real-trained auditor. A value above 1 means it beat the real-trained auditor. A negative value means it did worse than the null. The denominator is the real model’s improvement over the null; when that improvement is negligible the lineage is not informative and is omitted from the median. The three auditors are CatBoost [7], ordinary linear regression, and an MLP. Each reported median is the median across lineages of the median of three sample seeds.

MFS-v2 is a different object. With component $c_i \in [0, 1]$, weight $w_i > 0$, $\epsilon = 10^{-6}$, and $W = \sum_i w_i$,

$$\text{MFS-v2} = 100 \exp\left(\frac{1}{W} \sum_i w_i \ln(\epsilon + c_i)\right) \quad (3)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside $[0, 1]$. Otherwise the scalar is null. The weights are utility transfer 0.30, driver fidelity 0.20, distribution fidelity 0.20, structure fidelity 0.15, coverage realism 0.10, and compactness 0.05. Privacy’s soft weight is zero, so a privacy failure removes the score through a gate and does not move (3) directly. Utility transfer inside (3) is a clamped one-sided 95% lower bound on gold retention. It is not the median of (2). Compactness is $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$. Crossing 10,240 bytes fails the byte gate, so the scalar stays null.

Table I applies (3) to six invented components (0.90, 0.80, 0.85, 0.70, 0.75, 0.95). The resulting 82.40 is arithmetic, not a measurement. The second row is the same components with the byte gate failed. The implementation returns null. No artifact in this paper has a measured MFS-v2 value.

IV. PROTOCOL

The panel is the rights-cleared PMLB regression collection [8], [9]. Of 104 MIT-licensed entries, four were excluded because a source row appeared in both the official training and official test partitions. The remaining 100 use a grouped 80/20 fit and validation cut of the official training rows, seed 1729. Fit tables contain 72 to 800 rows. They are catalog coresets. Worker directories contain no official test file. Official tests stay sealed, and final evaluation is not authorized.

Fit seed 11 and sample seeds 101, 211, and 307 are the population actually measured. Sizes n and $4n$ are both stored; the tables below are $4n$. Native objectives are not ranked across methods. The Gaussian copula row is Copulas 0.14.1 `GaussianMultivariate`, selected by held-out mean log density. CTGAN and TVAE are the author implementation, selected by the mean of linear and MLP regression R^2 on validation rows [1]. ARF’s author `arfp` core has closed its FORDE log-density fits and has not yet been sampled on this

TABLE II
DENSITY BLOCK AT $4n$, FIT SEED 11. MEDIAN OF PER-LINEAGE THREE-SEED MEDIAN. STUDY CODE IS MARKED.

Auditor	Comparator	n	DOPE	Other
CatBoost	Gaussian copula	97	0.939876	0.655681
CatBoost	Chow–Liu, study	97	0.939876	0.585986
CatBoost	Indep. marginals, study	97	0.939876	−0.011971
Linear	Gaussian copula	92	0.994046	0.989300
Linear	Chow–Liu, study	92	0.994046	0.661218
Linear	Indep. marginals, study	92	0.994046	0.002056
MLP	Gaussian copula	75	1.061123	0.927569
MLP	Chow–Liu, study	75	1.061123	0.691645
MLP	Indep. marginals, study	75	1.061123	−0.027881

TABLE III
CHARGED BYTES AT $4n$. DENSITY METHODS USE 100 LINEAGES. CTGAN AND TVAE USE THE 22 NATIVE-SELECTED ARTIFACTS THAT PRODUCED A SIZE-4 SAMPLE, WHICH IS A WIDER SET THAN THE 21 INFORMATIVE RETENTION LINEAGES.

Method	Median bytes	Within cap
DOPE reference	1,895.5	98/100
Gaussian copula	5,464.5	62/100
Chow–Liu, study	29,865	31/100
Indep. marginals, study	2,549.5	94/100
CTGAN	741,834	0/22
TVAE	317,141.5	0/22

shared schedule [3]. Chow–Liu uses eight bins and Laplace smoothing 1.0 in study code. Independent marginals are a study reference with no author package.

V. RETENTION AND BYTES

Fig. 1 plots one point per method and lineage. DOPE sits to the left of the cap and near retention 1. Independent marginals are also small and sit near retention 0. The Gaussian copula and the study Chow–Liu tree spread across the cap. CTGAN and TVAE sit between about 10^5 and 10^6 bytes.

Table II is the paired density block. The CatBoost column repeats the medians already published for this profile. Linear and MLP come from the same paired records. The Gaussian copula is close to DOPE under the linear auditor (0.989300 versus 0.994046) and farther under CatBoost and the MLP. The study tree and the independent marginals stay below DOPE on all three auditors. Lineage counts differ because a lineage is omitted when that auditor’s real-versus-null gap is not informative. Median dataset-level differences are positive and the pre-registered paired test has not been run, so the comparison is inconclusive.

Table III counts artifacts at or under 10,240 charged bytes. Two of 100 DOPE artifacts exceed the cap (the largest is 22,233 bytes). Independent marginals pass the cap more often than the Chow–Liu study tree and still have retention near zero. Smallness alone is not the result.

Table IV is the neural block. It is not pooled with Table II. DOPE’s CatBoost median on these 21 lineages is 0.953366, CTGAN’s is −0.081259, and TVAE’s is 0.669033. The same direction appears for the linear auditor and the MLP, on 21 and 19 lineages. The ledger also records 525 scheduling cutoffs for the CTGAN/TVAE search. Those cutoffs are unfinished work, not method failures, and they are why this block cannot support a method-wide conclusion.

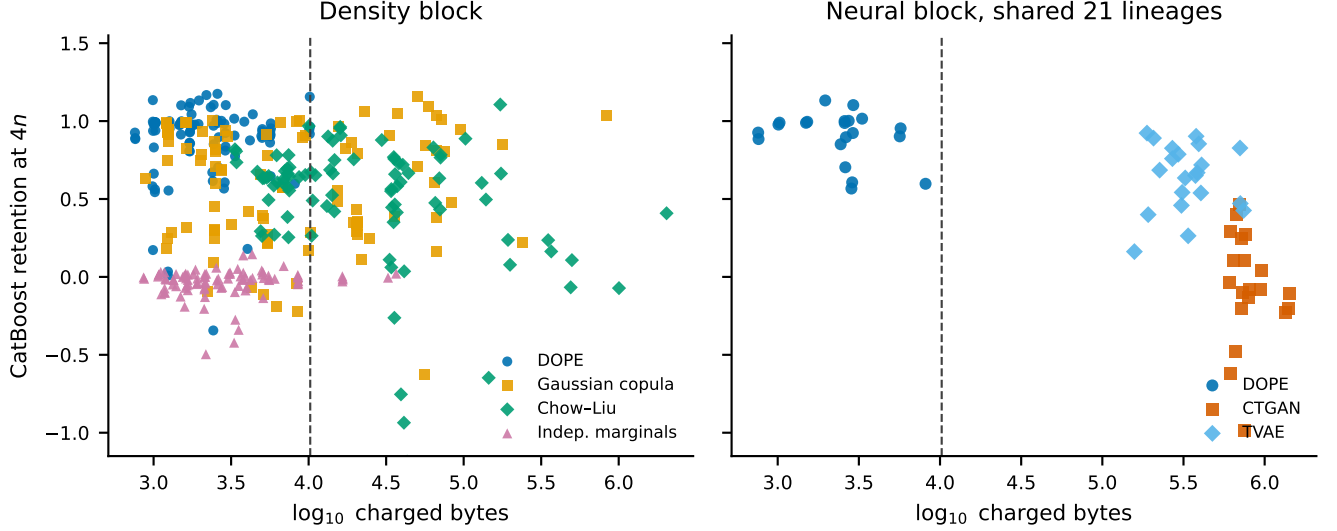


Fig. 1. CatBoost retention at $4n$ against charged artifact bytes. The dashed line is the 10,240-byte cap. The left panel is the density block. The right panel is the 21 lineages on which native-selected CTGAN, native-selected TVAE, and the DOPE reference all have an informative CatBoost median. The panels are not one ranking.

One shared lineage goes the other way. On 1193_BNG_lowbwt, native-selected TVAE retention is 1.029 at 293,868 bytes and the DOPE reference is 0.970 at 2,436 bytes. On 503_wind the same DOPE profile is 0.996 at 3,846 bytes and TVAE is 0.802 at 471,140 bytes.

VI. THE COMMON SCALAR

Table V is the MFS-v2 column required by Section III. No method has a complete component vector and a fully passed hard-gate map on this panel, so every scalar is null. Copying a retention number or a zero into this column would publish a value the contract refuses to emit. PTF-v1 is also null. Release-safe certification is null. The 100 lineages do not meet the production requirement of at least 30 informative groups under the final profile gate, and the five admission locks are not frozen.

VII. EVERY LINEAGE, NOT ONLY THE MEDIAN

Table II is the median of Appendix A. That appendix has one row for each of the 100 rights-cleared lineages: the PMLB name, the fit-row count, the feature count, the size- $4n$ byte charge, the L3 flag, and the three-seed median retention at n and at $4n$. A blank cell is an unavailable fit or an auditor that is not informative on all three sample seeds. It is not

TABLE IV
NEURAL BLOCK AT $4n$. CTGAN AND TVAE ARE NATIVE-SELECTED AUTHOR CODE. THIS LINEAGE SET IS NOT THE DENSITY BLOCK.

Auditor	Comparator	n	DOPE	Other
CatBoost	CTGAN	21	0.953366	-0.081259
CatBoost	TVAE	21	0.953366	0.669033
Linear	CTGAN	21	0.998972	-0.033499
Linear	TVAE	21	0.998972	0.795925
MLP	CTGAN	19	1.025597	-0.148653
MLP	TVAE	19	1.025597	0.752792

TABLE V
MFS-v2 AFTER NATIVE SELECTION. NULL MEANS THE SCALAR WAS NOT EMITTED.

Method	MFS-v2
DOPE reference	null
Gaussian copula	null
CTGAN	null
TVAE	null
ARF	null
TabDDPM	null
synthpop CART	null
Chow-Liu, study code	null
Indep. marginals, study code	null

a zero. Two displayed-profile fits are unavailable, 4544_GeographicalOriginalofMusic and 588_fri_c4_1000_100. 678_visualizing_environmental has a fit whose CatBoost group is not informative on all three seeds. The CatBoost median therefore uses 97 lineages, and the byte count uses 100.

Fig. 2 plots those medians against charged bytes for CatBoost. The same drawing for the linear auditor and the MLP is in the figure directory. The frame is $[-1.5, 1.6]$. Points outside it remain in the appendix; the panel title counts them. The dashed line is $\log_{10}(10240)$. Fig. 5 is the paired difference, displayed profile minus each native-selected density comparator, with the lineage name on the axis. The frame is $[-1.5, 1.5]$ for the same reason. These figures are descriptive. They are not a paired superiority test.

VIII. EIGHT THOUSAND STEPS ON THE SAME PARTITIONS

features12_steps8192 is the same residual with four times as many AdamW steps. It was already fit in the closed expansion round. Of the 100 lineages, 68 have a successful fit. Thirty failed before the fit started because of transport or prelaunch, one failed because a foreign GPU owner appeared,

features12_steps2048: catboost

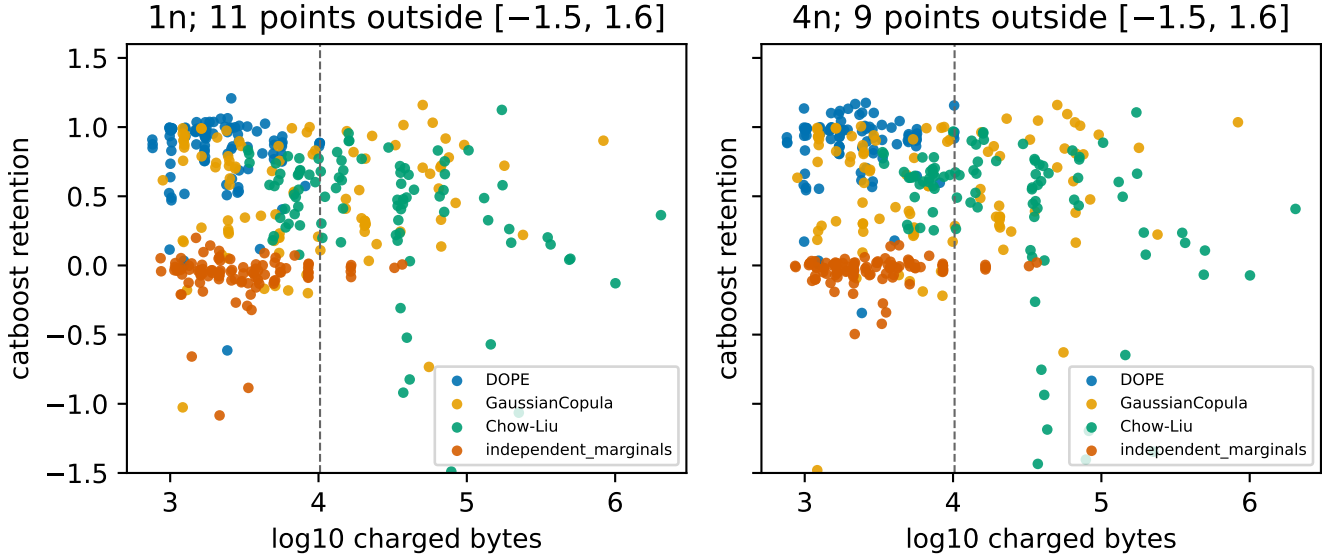


Fig. 2. CatBoost retention against charged bytes for every informative lineage of features12_steps2048 and the native-selected density comparators. The neural block is not on this figure.

and one failed the charged-artifact cap. Sample evaluation of that round has 408 measured cells and 192 fit-unavailable cells for this profile. Sixty-eight of the 69 artifacts with a byte charge are within 10,240 bytes. The largest charged artifact is 19,041 bytes.

On the lineages where both budgets have a complete three-seed group, the CatBoost medians are 0.925519 at 2,048 steps and 0.970348 at 8,192 steps (68 lineages; median paired difference 0.034349). The linear medians are 0.990163 and 0.986519 (64 lineages; median paired difference -0.000283). The MLP medians are 1.065260 and 1.056573 (52 lineages; median paired difference 0.004514). The extra steps are a budget check. They do not replace features12_steps2048 as the displayed model, and they are not a superiority test. Appendix B lists all 100.

The scored fits did not retain a per-step loss. A logged replay of both budgets, written to a new directory and not substituted for those artifacts, records the AdamW training loss and the training-derived validation loss at every step. The curves are the hidden-basis objective, before the closed-form readout refit. All 100 lineages finished both budgets, and every recorded loss is finite. The median training loss starts at 0.321246. It ends at 0.002745 after 2,048 steps and at 0.000763 after 8,192 steps. The median validation loss starts at 0.270883 and ends at 0.004433 and 0.001813. Figure 6 and Figure 7 draw the median and the band from the 10th to the 90th percentile. losses.png is not that series. The lower hidden-basis loss at 8,192 steps does not replace the displayed profile.

IX. BEYONDARENA IS A SEPARATE CORPUS

BeyondArena is the pinned TabArena/BeyondArena snapshot 2ecfe882ccfb814fc27c4de10a64ceefd5d7655c in

s3://veox-jopedime-datasets-use2/
benchmarks/BeyondArena/.

The index lists 142 source families, 19,676,576 rows, and 3,722 official folds. The task mix is 73 binary, 44 regression, and 25 multiclass. The split regimes are 103 IID, 21 temporal, and 18 grouped. Every family is stored once under raw/BeyondArena/ at that revision. The research prefix research/BeyondArena/ holds later experiment handoffs, not a second copy of the tables. Appendix C names all 142.

The fit lock takes at most 12 binary or regression families whose recorded license is a redistribution grant, labels each with the published split regime, and round-robins IID, grouped, and temporal in hash order of the source-family id. Multiclass families and licenses without that grant stay out. None of the 12 is one of the 100 PMLB lineages. The locked names are airfoil_self_noise, early_learning_predictors, lending_club, heart_disease_va_long_beach, musk, kick, qsar_biodeg, telemonitoring_parkinsons, biomedical_voice_measurements, hotel_booking_demand, wine_world_cost, video_transcoding_time_prediction, and garments_worker_productivity.

Official fold 0 of repeat 0 is the fit source. An 80/20 grouped cut of that official training fold, seed 1729, supplies the fit and the validation loss. The official test rows stay in the evaluator directory. Four families, early_learning_predictors, lending_club, kick, and wine_world_cost, exceed the 2,000-feature projection cap. lending_club still does after 14 all-missing training columns are dropped. heart_disease_va_long_beach,

features12_steps2048: linear

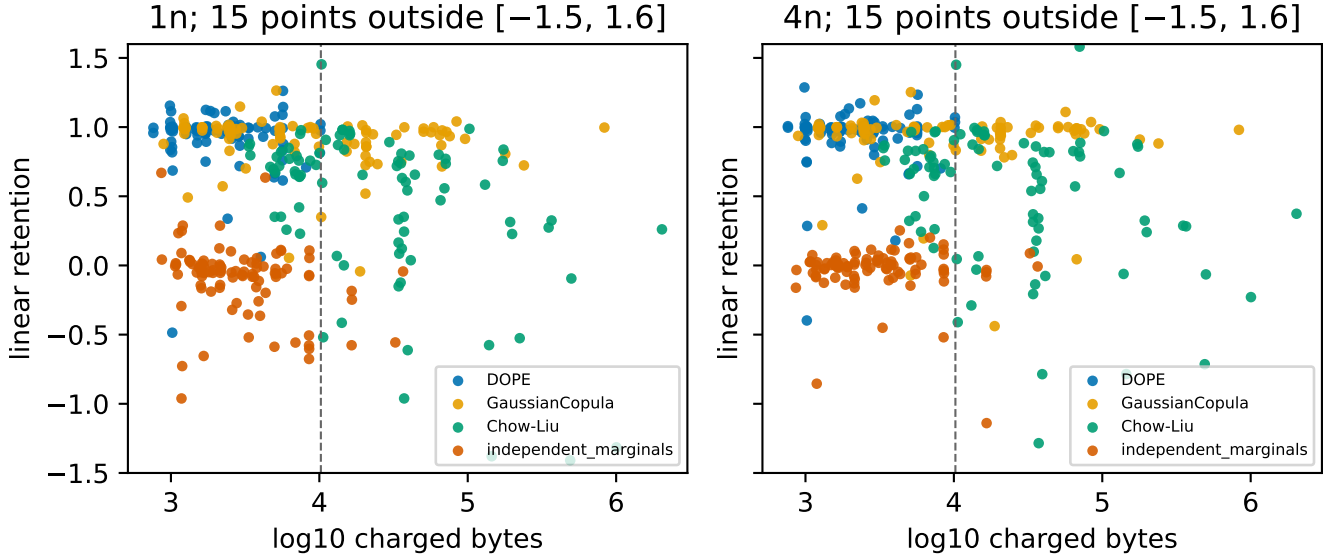


Fig. 3. Linear retention against charged bytes, same lineages and frame as Fig. 2.

TABLE VI

BEYONDARENA FAMILIES THAT PREPARED. BYTES ARE THE CHARGED ARTIFACT. THE LOSS IS THE FINAL ADAMW HIDDEN-BASIS VALIDATION LOSS, BEFORE THE READOUT REFIT.

family	fit rows	2,048 B	8,192 B	val 2,048	val 8,192
airfoil_self_noise	802	804	818	0.0101	0.0045
musk	3,738	9,273	9,350	0.1456	0.0903
telemonitoring	257	1,852	1,890	0.0333	0.0190
video_transcoding	35,918	5,054	5,071	0.0025	0.0020
garments_productivity	939	2,317	2,372	0.0277	0.0251

qsar_biodeg, and hotel_booking_demand have an identical source row on both sides of the official fold, the same overlap rule that excluded four PMLB lineages. The other five families completed both budgets. Every artifact is inside the 10,240-byte cap. Table VI records the charged bytes and the final hidden-basis validation loss. No retention auditor was run on this block, so it has no retention column, and it is not pooled with the 100 or with the 21-lineage neural block. (3) stays null.

X. WHAT THIS PANEL DOES NOT ESTABLISH

The fits use coresets of 72–800 rows, one fit seed, and a training-derived validation cut. They do not establish behavior on the full tables used by TabDDPM, a five-seed campaign, production PTF-v1, or a release decision. ARF sampling and the TabDDPM run on these 100 lineages are still ahead. A closed refinement-validation round accounts for 1,200 logical cells, 816 measured and 384 unavailable, and it likewise leaves (3) null. Its public report is not yet issued: each measured sample directory contains two unlisted scratch directories, so the receipt inventory check fails closed. Nothing in that failure opens an official test or assigns a fitness score.

When a later round measures the six components, Table V is the column to fill, and only through the implementation of (3). A paired interval can be added only after the protocol’s paired test is computed. Until then the public statement is the one in the abstract: the residual is specified, retention and bytes are measured on the blocks above, and the common scalar is null.

LEDGER

The medians, lineage counts, byte ranges, and null flags are the 5 October 2026 entries in research/benchmark/RESULTS_STATUS.md, taken from the density-matched and SDV-matched population validation files and from research/benchmark/results/s3-lineage-record.json. Fitness weights are production/kpi-contract.json. The displayed profile remains features12_steps2048.

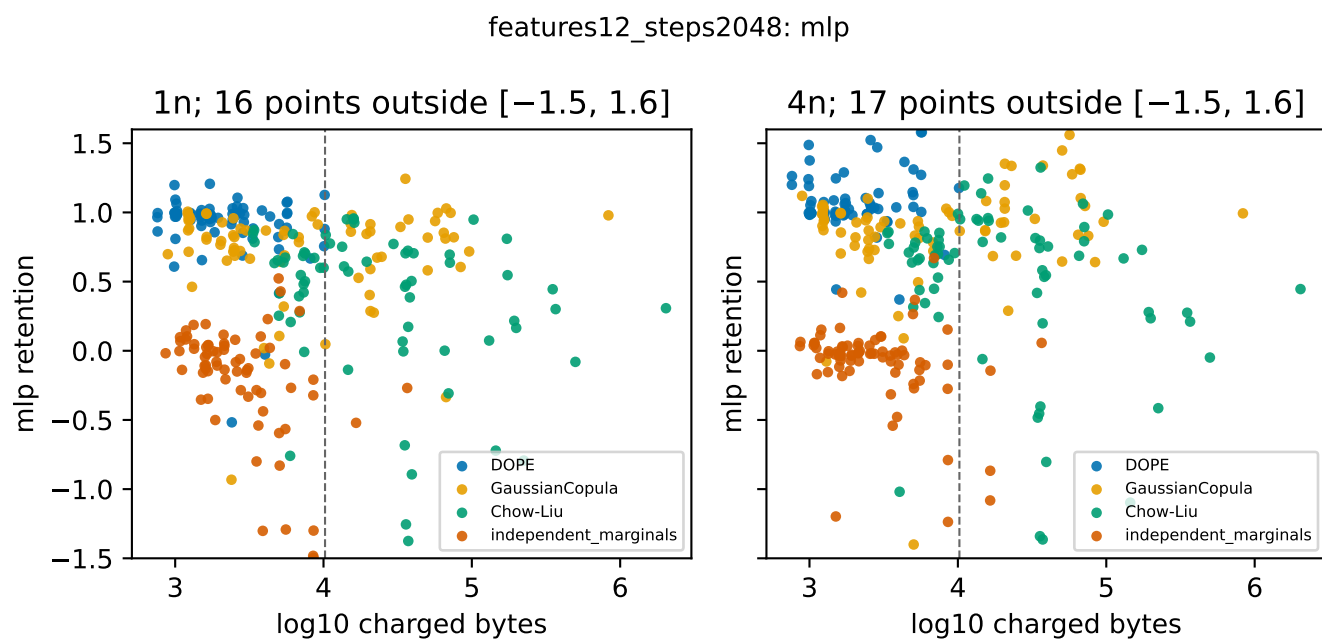


Fig. 4. MLP retention against charged bytes, same lineages and frame as Fig. 2.

CatBoost retention difference at 4n, features12_steps2048

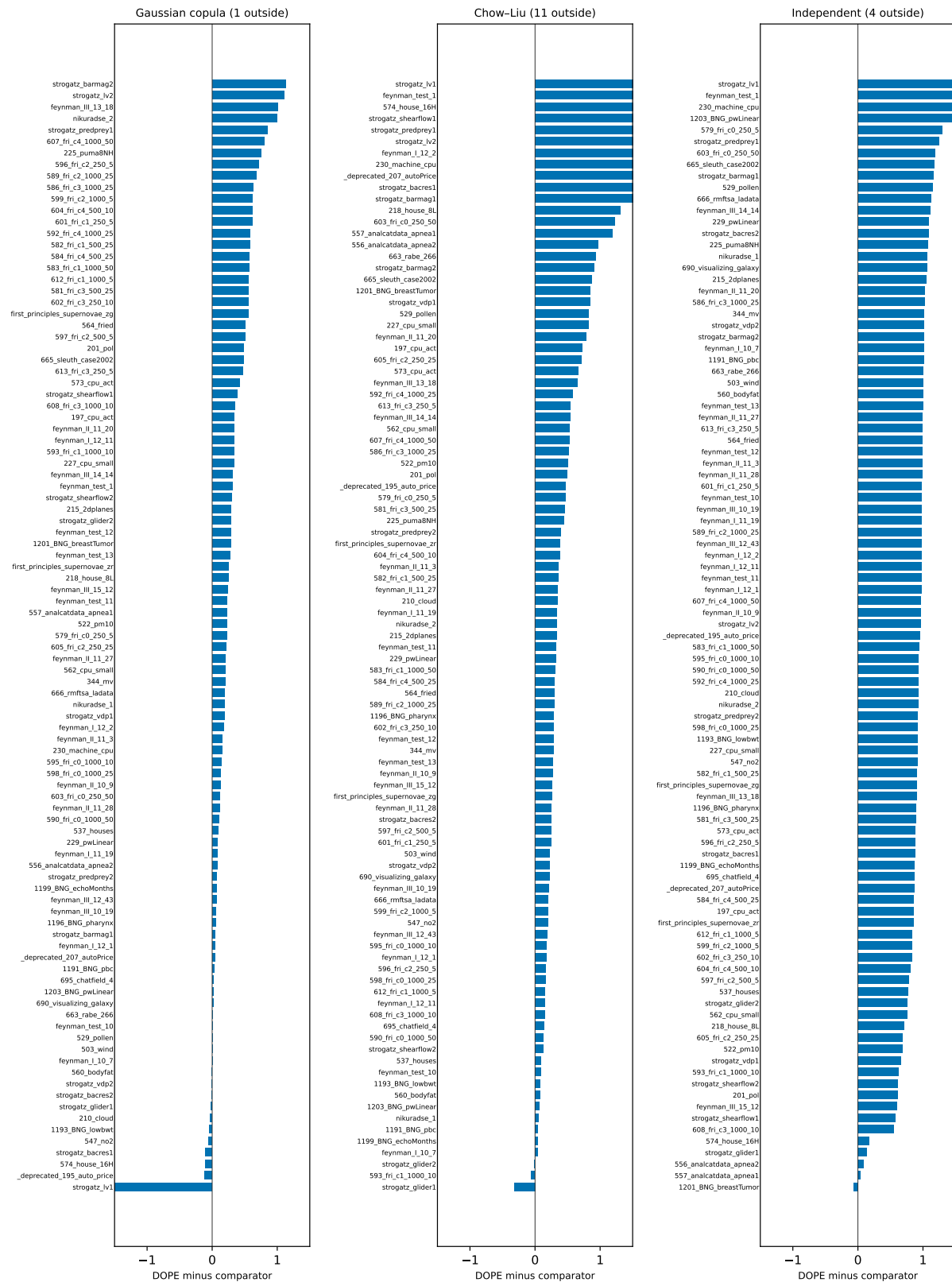


Fig. 5. Paired CatBoost difference at 4n. Each bar is one informative lineage. Differences outside $[-1.5, 1.5]$ are counted in the panel title and listed in Appendix A.

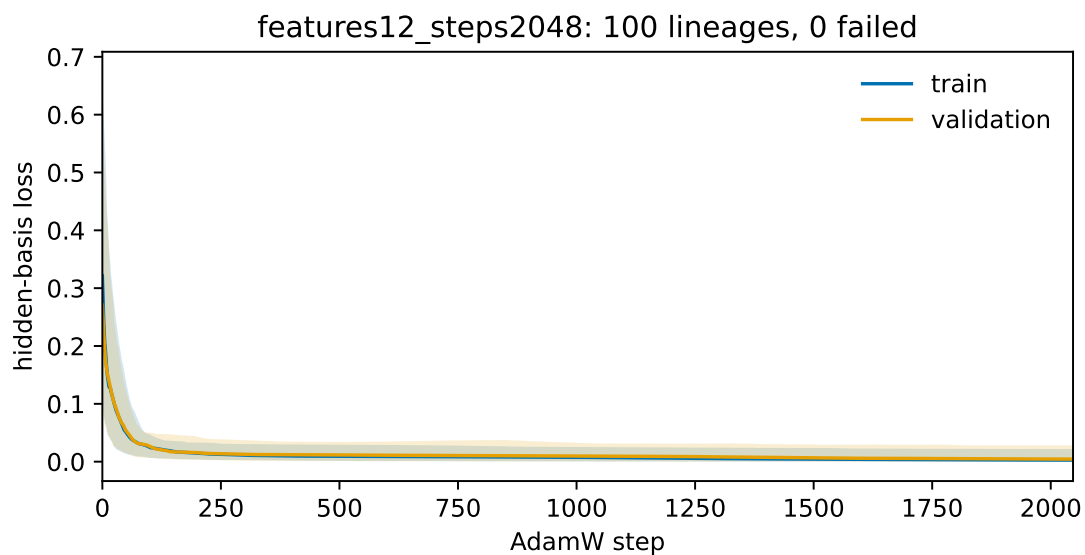


Fig. 6. AdamW hidden-basis loss at 2,048 steps. The line is the median of 100 lineages. The band is the 10th to the 90th percentile, before the readout refit.

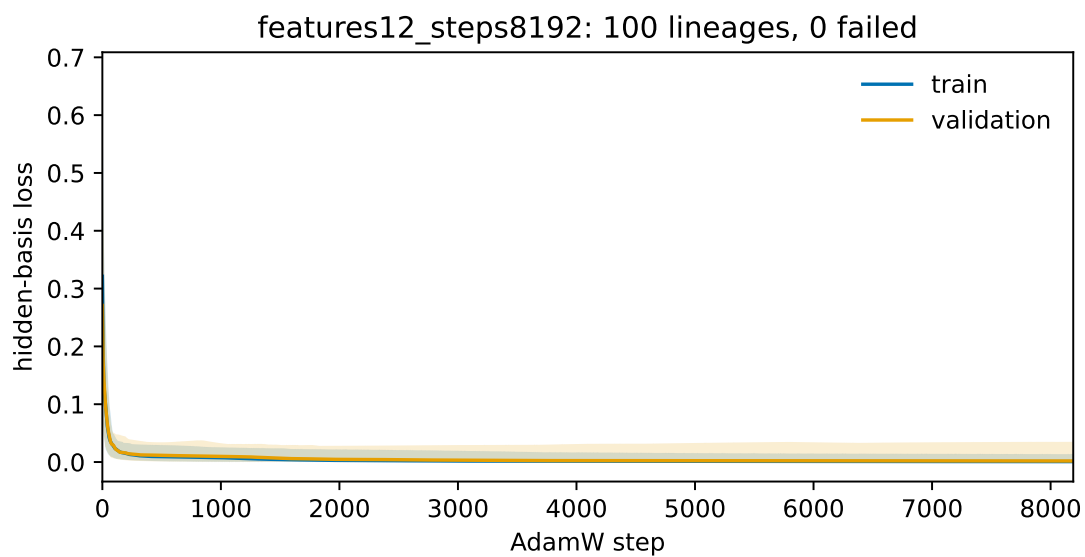


Fig. 7. AdamW hidden-basis loss at 8,192 steps, same 100 lineages and same band. This budget is not the displayed model.

APPENDIX A

DISPLAYED PROFILE, ALL 100 LINEAGES

TABLE VII: Every rights-cleared lineage at the displayed profile features 12_steps2048. A blank retention means the three-seed group is not informative or the fit is unavailable. Bytes and the L3 flag are the size-4n charge. MFS-v2 is not in this table.

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
02a45777900441ce	215_2dplanes	800	10	2,583	pass	0.9930	1.0001	0.9990	1.0477
047e62b52a3f4ede	695_chatfield_4	156	12	3,311	pass	1.0041	1.0155	1.0048	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	762	pass	0.8753	0.8846	0.9959	1.2627
0561d673cb73c27e	666_rmftsa_ladata	339	10	2,921	pass	1.0095	1.1026	1.0338	1.1394
060b9fff8092db17	589_fri_c2_1000_25	667	25	5,717	pass	0.8893	0.9534	0.9911	1.5794
06b0312a585b436e	1196_BNG_pharynx	800	10	2,631	pass	0.8601	0.8964	1.0458	1.7049
0779d3fd5b3133a2	nikuradse_2	241	1	758	pass	0.9096	0.9261	2.1723	2.2082
07f38946316c039b	593_fri_c1_1000_10	667	10	2,876	pass	0.5650	0.6068	0.9617	1.6005
0a8428ef25a69175	537_houses	800	8	2,609	pass	0.6257	0.7036	1.0150	1.0256
0d482c1dd7cc6592	210_cloud	72	5	1,844	pass	1.0374	0.9772	1.0944	1.0239
0db59fd2579600d0	665_sleuth_case2002	98	6	1,962	pass	1.0371	1.1324	1.1345	—
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,183	pass	0.8097	0.8761	0.9173	—
167578f5deba4977	547_no2	333	7	2,443	pass	0.8073	0.8513	1.0097	0.9898
19f2c3c2d67defd7	feynman_II_11_27	800	4	1,494	pass	0.9704	0.9890	0.9960	1.0048
19f4780b53b3fa41	678_visualizing_environmental	74	3	1,515	pass	—	—	—	0.4431
1a4456d1cc71a26a	feynman_test_1	800	7	2,191	pass	1.0608	1.1671	0.9816	—
1a803b27c2400dc4	feynman_III_10_19	800	4	1,520	pass	0.9882	0.9952	0.9942	0.9965
1cb97b1d7314d43d	229_pwLinear	133	10	2,587	pass	0.8417	0.9878	0.9104	—
205f4b4982b9ea49	608_fri_c3_1000_10	667	10	2,845	pass	0.5263	0.5671	1.0451	0.8182
26eff843be35d831	581_fri_c3_500_25	333	25	5,647	pass	0.7602	0.9020	1.1297	1.5792
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,904	pass	0.8974	0.9234	0.9962	1.0250
297b7688a045c583	603_fri_c0_250_50	167	50	10,171	pass	0.8848	1.1562	1.0700	2.6883
2c8d0985f3557d2b	feynman_II_11_28	800	2	1,026	pass	0.9831	0.9899	1.0031	0.9976
30ca2887550f0d8d	201_pol	800	26	8,145	pass	0.5742	0.5976	0.7015	0.6923
3408a6a513c627a3	strogatz_barmag1	267	2	1,011	pass	0.9750	0.9775	0.9969	1.0058
35a9aa9b88f0e4d1	344_mv	800	10	2,737	pass	0.9985	1.0016	0.9662	1.0347
3d3321e27d14f9aa	599_fri_c2_1000_5	667	5	1,707	pass	0.8359	0.8611	0.9668	1.6797
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,712	pass	0.8671	0.9276	0.8599	—
46b25db9396c8209	227_cpu_small	800	12	3,327	pass	0.8569	0.9072	0.9018	0.9364
46c5a59e5ae42037	1193_BNG_lowbw	800	9	2,436	pass	0.9635	0.9703	1.0030	0.9871
471f11cc2d38853a	feynman_test_11	800	4	1,481	pass	0.9645	0.9760	0.9975	0.9826
47c581af5de1136b	582_fri_c1_500_25	333	25	5,670	pass	0.7510	0.9088	0.7173	—
48b7a3bf81e5f3e5	605_fri_c2_250_25	167	25	5,692	pass	0.4046	0.6465	1.9138	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,713	pass	0.7550	0.8055	0.9814	1.8381
4aa84fff8ad0cd26	feynman_III_13_18	800	4	1,501	pass	0.8748	0.9137	0.8850	0.9348
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,237	pass	-0.0032	0.0086	—	—
54b221668673164c	feynman_III_14_14	800	5	1,726	pass	1.0346	1.1164	1.0011	1.0611
552665868f8f72ed	strogatz_vdp2	267	2	989	pass	0.9945	0.9963	0.9960	0.9969
585fef2a1cb6b33f	strogatz_lv2	267	2	1,018	pass	0.8286	0.9859	0.7502	—

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
642738a361f1c63a	feynman_I_12_1	800	2	1,032	pass	0.9892	0.9910	0.9892	0.9933
64c64f0f9983c48f	strogatz_bacres2	267	2	998	pass	0.9793	0.9832	0.9783	0.9862
654566a1f91408aa	4544_GeographicalOriginalofMusic	706	117	22,233	fail	—	—	—	—
693890aae9dc860b	first_principles_supernovae_zg	162	1	761	pass	0.8492	0.8908	1.0066	1.1997
6cacb11611345a24	feynman_I_11_19	800	6	1,947	pass	0.9524	0.9812	0.9780	0.9800
7291b347fe0238fb	feynman_II_11_20	800	5	1,757	pass	0.9928	1.0432	1.0016	1.0674
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,021	pass	0.4900	0.5614	-0.3978	—
7a3d123dd555110d	598_fri_c0_1000_25	667	25	5,668	pass	0.8613	0.9166	1.0046	1.0034
7c8943b160464add	601_fri_c1_250_5	167	5	1,719	pass	0.7859	0.8552	1.0773	—
7e5171baa1b1fb1	feynman_I_12_11	800	5	1,718	pass	0.9267	0.9320	1.0015	0.9860
80589368fa78f21a	218_house_8L	800	8	2,392	pass	0.5835	0.6708	0.9616	1.1040
8164950b155f7f56	feynman_I_12_2	800	4	1,511	pass	0.9702	0.9986	0.9485	0.9753
85a31ae409944a1b	feynman_II_10_9	800	3	1,269	pass	0.9330	0.9508	0.9589	0.9515
8d859cfa7b51a351	feynman_III_15_12	800	3	1,248	pass	0.5172	0.5542	1.0228	1.0749
923483635406b213	1203_BNG_pwLinear	800	10	2,581	pass	1.2079	1.1750	0.9941	1.5230
9312d813229df035	690_visualizing_galaxy	216	4	1,458	pass	0.9725	0.9755	0.9853	1.0870
94a00ea67f49bf84	596_fri_c2_250_5	167	5	1,698	pass	0.7546	0.8072	—	—
94bd4520b8ceb3f0	feynman_test_12	800	5	1,747	pass	0.9688	0.9804	0.9913	1.0067
956d34f13e7af830	586_fri_c3_1000_25	667	25	5,701	pass	0.9325	0.9849	1.2339	2.1457
9a10c15885022034	_deprecated_195_auto_price	106	15	5,025	pass	0.8717	0.9485	0.9598	1.3107
a04f964bc53281e0	strogatz_lv1	267	2	985	pass	-3.8784	-3.4212	—	—
a23f4bd626b76590	feynman_test_13	800	5	1,740	pass	0.9178	0.9605	0.9900	1.0991
a32a69e7c701eb14	588_fri_c4_1000_100	667	100	19,019	fail	—	—	—	—
a5ba682572f53add	602_fri_c3_250_10	167	10	2,900	pass	0.5355	0.8109	0.7988	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9781	0.9933	0.9967	1.0521
a97065df07a47eb0	592_fri_c4_1000_25	667	25	5,655	pass	0.8791	0.9340	0.9578	1.2714
ab279eded3b3c52b	574_house_16H	800	16	4,035	pass	0.1181	0.1796	0.1795	0.3705
aebf5d861a8365b4	522_pm10	333	7	2,409	pass	0.5082	0.6149	0.4125	2.3076
b6adf79118a58cea	562_cpu_small	800	12	3,353	pass	0.6968	0.7747	0.9834	0.9307
b78d323566930bd7	583_fri_c1_1000_50	667	50	10,098	pass	0.8607	0.9444	0.9960	1.6211
b7f9774e2142a758	strogatz_shearflow2	267	2	1,019	pass	0.4707	0.5439	0.7450	1.8955
bc85856c012bcc1b	607_fri_c4_1000_50	667	50	10,209	pass	0.8856	0.9640	3.9394	—
c107ee39eb333a9d	584_fri_c4_500_25	333	25	5,638	pass	0.8062	0.8619	0.9417	—
c30ff9405c916718	579_fri_c0_250_5	167	5	1,692	pass	0.9475	1.0923	0.9961	0.9783
c4f8e81b70d33a13	590_fri_c0_1000_50	667	50	10,180	pass	0.8839	0.9172	1.0359	1.1762
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0356	1.1012	0.9717	1.2459
cb5bf66ef2ee09ed	nikuradse_1	241	2	992	pass	0.9346	0.9373	0.9997	1.1909
cba8393879afd1dc	230_machine_cpu	140	6	2,338	pass	1.0026	1.0891	1.1718	—
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,003	pass	0.6382	0.6625	1.0720	1.3756
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7352	0.8309	0.9014	0.9826
cee6f811d0bf199d	feynman_II_11_3	800	5	1,728	pass	0.9385	0.9666	0.8950	1.0149
d255d3a2e1c8dbf4	feynman_III_12_43	800	2	1,028	pass	0.9920	0.9935	0.9966	0.9940
d52bfd2b6ab1c850	feynman_I_10_7	800	3	1,250	pass	0.9920	0.9932	0.9940	0.9959
d9a3408a840c9f48	560_bodyfat	168	14	4,364	pass	1.0247	1.0436	0.9930	1.3651
da290fae0d6c072f	feynman_test_10	800	3	1,267	pass	0.9955	0.9990	0.9990	1.0015
da78606950f100cd	1199_BNG_echoMonths	800	9	2,544	pass	0.9962	1.0086	0.9655	1.0694
dd16ac872ffb8673	strogatz_glider1	267	2	991	pass	0.1145	0.1723	0.9123	—
de2868ac837699fd	564_fried	800	10	2,861	pass	0.9477	0.9829	0.9543	1.4707
e08f0b9931f93686	225_puma8NH	800	8	2,439	pass	1.0639	1.1043	0.9951	2.8498
e18350fb51a9d1b0	strogatz_barmag2	267	2	1,027	pass	0.8897	0.9399	0.2855	—
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,238	pass	0.0339	0.0341	—	—
e68fdd96cc63271b	573_cpu_act	800	21	4,954	pass	0.8080	0.9045	0.6593	1.1465
ed2efb82fce4b098	strogatz_vdp1	267	2	984	pass	0.5228	0.5812	1.2867	1.2401
edc9d41a298fd60	612_fri_c1_1000_5	667	5	1,705	pass	0.8224	0.8596	0.9670	1.2891
efd53e9c0f36d608	_deprecated_207_autoPrice	106	15	5,017	pass	0.9020	0.9357	1.0899	1.0398
f0a3d6881458f990	604_fri_c4_500_10	333	10	2,913	pass	0.6868	0.8360	0.8552	—
f24e9383331c43e2	503_wind	800	14	3,846	pass	0.9690	0.9962	1.0017	1.0367
f652ea96eff62654	strogatz_predprey1	267	2	989	pass	0.9964	1.1345	1.0218	1.4876
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9876	0.9847	0.9718	1.0815
fc3812567b5fdd64	197_cpu_act	800	21	5,011	pass	0.7680	0.8922	0.6669	1.1253
fe6d7ac3fd3129e	1201_BNG_breastTumor	800	9	2,427	pass	-0.6141	-0.3437	—	—

APPENDIX B

SUFFICIENCY BUDGET, ALL 100 LINEAGES

TABLE VIII: Every rights-cleared lineage at the pre-registered sufficiency budget features12_steps8192. This profile is not adopted as the displayed model. A blank retention means the three-seed group is not informative or the fit is unavailable.

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
02a45777900441ce	215_2dplanes	800	10	—	—	—	—	—	—
047e62b52a3f4ede	695_chatfield_4	156	12	3,353	pass	1.0106	1.0287	1.0110	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	750	pass	0.9372	0.9454	1.0346	1.3271
0561d673cb73c27e	666_rmftsa_ladata	339	10	—	—	—	—	—	—
060b9fff8092db17	589_fri_c2_1000_25	667	25	5,734	pass	0.9237	0.9886	1.0258	1.5430
06b0312a585b436e	1196_BNG_pharynx	800	10	2,681	pass	0.8188	0.8768	1.0632	0.7968
0779d3fd5b3133a2	nikuradse_2	241	1	—	—	—	—	—	—
07f38946316c039b	593_fri_c1_1000_10	667	10	2,869	pass	0.9780	1.0222	0.9929	1.8097
0a8428ef25a69175	537_houses	800	8	2,636	pass	0.7243	0.7841	1.0073	1.0334
0d482c1dd7cc6592	210_cloud	72	5	1,855	pass	0.9561	0.8815	1.0056	0.9224

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
0db59fd2579600d0	665_sleuth_case2002	98	6	—	—	—	—	—	—
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,218	pass	0.8069	0.8576	0.8783	—
167578f5deba4977	547_no2	333	7	2,474	pass	0.8821	0.9276	0.9850	1.0276
19f2c3c2d67defd7	feynman_II_11_27	800	4	—	—	—	—	—	—
19f4780b53b3fa41	678_visualizing_environmental	74	3	—	—	—	—	—	—
1a4456d1cc71a26a	feynman_test_1	800	7	2,185	pass	1.1162	1.3026	0.6721	—
1a803b27c2400dc4	feynman_III_10_19	800	4	1,522	pass	0.9698	0.9777	0.9732	0.9807
1cb97b1d7314d43d	229_pwLinear	133	10	2,598	pass	0.8275	1.0471	0.9727	—
205f4b4982b9ea49	608_fri_c3_1000_10	667	10	—	—	—	—	—	—
26eff843be35d831	581_fri_c3_500_25	333	25	5,655	pass	0.8660	0.9960	1.0900	1.6067
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,913	pass	0.9645	1.0058	1.0029	1.1269
297b7688a045c583	603_fri_c0_250_50	167	50	10,183	pass	0.8257	1.1292	1.0240	2.8999
2c8d0985f3557d2b	feynman_II_11_28	800	2	—	—	—	—	—	—
30ca2887550f0d8d	201_pol	800	26	8,215	pass	0.6214	0.6436	0.7350	0.6603
3408a6a513c627a3	strogatz_barmag1	267	2	—	—	—	—	—	—
35a9aa9b88f0e4d1	344_mv	800	10	—	—	—	—	—	—
3d3321e27d14f9aa	599_fri_c2_1000_5	667	5	1,713	pass	0.9820	1.0044	1.0441	1.6148
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,708	pass	0.8605	0.9611	0.9427	—
46b25db9396c8209	227_cpu_small	800	12	3,338	pass	0.7033	0.8730	0.9142	0.9908
46c5a59e5ae42037	1193_BNG_lowbwt	800	9	2,468	pass	0.8393	0.8390	0.8798	0.8103
471f11cc2d38853a	feynman_test_11	800	4	1,472	pass	0.9340	0.9481	0.9730	0.9578
47c581af5de1136b	582_fri_c1_500_25	333	25	5,676	pass	0.8232	0.9820	0.8086	—
48b7a3bf81e5f3e5	605_fri_c2_250_25	167	25	5,696	pass	0.2608	0.5247	1.4268	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,715	pass	0.9578	1.0001	1.0224	1.3200
4aa84fff8ad0cd26	feynman_III_13_18	800	4	—	—	—	—	—	—
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,246	pass	0.6556	0.6654	—	—
54b221668673164c	feynman_III_14_14	800	5	1,724	pass	1.0175	1.1226	1.0021	1.0727
552665868f8f72ed	strogatz_vdp2	267	2	—	—	—	—	—	—
585fef2a1cb6b33f	strogatz_lv2	267	2	1,021	pass	0.9239	1.0451	0.8041	—
642738a361f1c63a	feynman_I_12_1	800	2	1,034	pass	0.9936	0.9961	0.9993	0.9989
64c64f0f9983c48f	strogatz_bacres2	267	2	996	pass	0.9743	0.9826	0.9816	0.9923
654566a1f91408aa	4544_GeographicalOriginalofMusic	706	117	—	—	—	—	—	—
693890aae9dc860b	first_principles_supernovae_zg	162	1	766	pass	0.9415	0.9642	1.0277	1.1876
6cacb11611345a24	feynman_I_11_19	800	6	1,952	pass	0.9911	1.0176	1.0047	1.0089
7291b347fe0238fb	feynman_II_11_20	800	5	—	—	—	—	—	—
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,023	pass	0.4860	0.5973	-4.3352	—
7a3d123dd55110d	598_fri_c0_1000_25	667	25	5,672	pass	0.8570	0.9206	1.0010	1.0168
7c8943b160464add	601_fri_c1_250_5	167	5	—	—	—	—	—	—
7e51711baa1b1fb1	feynman_I_12_11	800	5	1,720	pass	0.9780	0.9927	0.9910	1.0559
80589368fa78f21a	218_house_8L	800	8	2,410	pass	0.6661	0.7528	0.6830	1.0728
8164950b155f7f56	feynman_I_12_2	800	4	—	—	—	—	—	—
85a31ae409944a1b	feynman_II_10_9	800	3	1,264	pass	0.9803	0.9961	0.9881	1.0061
8d859cfa7b51a351	feynman_III_15_12	800	3	1,280	pass	0.7428	0.8572	0.8771	1.2306
923483635406b213	1203_BNG_pwLinear	800	10	—	—	—	—	—	—
9312d813229df035	690_visualizing_galaxy	216	4	1,470	pass	0.9770	0.9875	0.9843	1.0661
94a00ea67f49bf84	596_fri_c2_250_5	167	5	1,718	pass	0.8262	0.8934	—	—
94bd4520b8ceb3f0	feynman_test_12	800	5	1,745	pass	0.9677	0.9845	0.9937	1.0157
956d34f13e7af830	586_fri_c3_1000_25	667	25	—	—	—	—	—	—
9a10c15885022034	_deprecated_195_auto_price	106	15	5,038	pass	0.8965	0.9544	0.9901	1.3165
a04f964bc53281e0	strogatz_lv1	267	2	986	pass	-3.0531	-2.4502	—	—
a23f4bd626b76590	feynman_test_13	800	5	—	—	—	—	—	—
a32a69e7c701eb14	588_fri_c4_1000_100	667	100	19,041	fail	—	—	—	—
a5ba682572f53add	602_fri_c3_250_10	167	10	2,902	pass	0.7774	0.9773	0.9471	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9849	0.9996	1.0018	1.0573
a97065df07a47eb0	592_fri_c4_1000_25	667	25	—	—	—	—	—	—
ab279eded3b3c52b	574_house_16H	800	16	4,076	pass	0.0805	0.1211	-0.7472	-0.3199
aebf5d8618a365b4	522_pm10	333	7	2,450	pass	0.6372	0.7959	0.6203	3.4124
b6ad1f79118a58cea	562_cpu_small	800	12	—	—	—	—	—	—
b78d323566930bd7	583_fri_c1_1000_50	667	50	10,098	pass	0.8817	0.9737	0.8733	0.9465
b7f9774e2142a758	strogatz_shearflow2	267	2	1,022	pass	0.7388	0.8339	-0.4873	1.5681
bc85856c012bcc1b	607_fri_c4_1000_50	667	50	—	—	—	—	—	—
c107ee39eb333a9d	584_fri_c4_500_25	333	25	5,641	pass	0.8307	0.9024	0.9623	—
c30ff9405c916718	579_fri_c0_250_5	167	5	1,700	pass	0.9081	1.0750	0.9949	0.9830
c4f8e81b70d33a13	590_fri_c0_1000_50	667	50	—	—	—	—	—	—
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0619	1.1198	0.9983	1.2801
cb5bf66ef2ee09ed	nikuradse_1	241	2	990	pass	0.9662	0.9677	0.9970	1.1953
cba8393879afd1dc	230_machine_cpu	140	6	—	—	—	—	—	—
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,004	pass	0.5396	0.6099	1.1474	1.2017
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7807	0.8568	0.8858	0.9570
cee6f811d0bf199d	feynman_II_11_3	800	5	—	—	—	—	—	—
d255d3a2e1c8dbf4	feynman_III_12_43	800	2	1,023	pass	0.9947	0.9964	0.9984	0.9967
d52bfd2b6ab1c850	feynman_I_10_7	800	3	1,248	pass	0.9978	0.9988	0.9994	1.0005
d9a3408a840c9f48	560_bodyfat	168	14	—	—	—	—	—	—
da290fae0d6c072f	feynman_test_10	800	3	1,257	pass	0.9681	0.9718	0.9737	0.9744
da78606950f100cd	1199_BNG_echoMonths	800	9	2,578	pass	0.9473	0.9673	0.9447	1.0028
dd16ac872ffb8673	strogatz_glider1	267	2	—	—	—	—	—	—
de2868ac837699fd	564_fried	800	10	2,864	pass	0.9826	1.0201	0.9548	1.5094
e08f0b9931f93686	225_puma8NH	800	8	2,440	pass	1.0168	1.0664	0.9433	2.7529
e18350fb51a9d1b0	strogatz_barmag2	267	2	—	—	—	—	—	—
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,241	pass	0.6282	0.6707	—	—
e68fdd96cc63271b	573_cpu_act	800	21	4,958	pass	0.8507	0.9689	0.5661	1.1177
ed2efb82fce4b098	strogatz_vdp1	267	2	991	pass	0.5397	0.6472	-0.1705	1.1154
edc9d41a298fdff60	612_fri_c1_1000_5	667	5	—	—	—	—	—	—
efd53e9cf36d608	_deprecated_207_autoPrice	106	15	5,024	pass	0.9035	0.9215	1.0781	1.0481

id	name	rows	feat.	bytes	L3	CB n	CB $4n$	lin $4n$	MLP $4n$
f0a3d6881458f990	604_fri_c4_500_10	333	10	2,918	pass	0.8895	1.0064	1.0421	—
f24e9383331c43e2	503_wind	800	14	3,877	pass	0.9992	0.9953	0.9918	1.0034
f652ea96eff62654	strogatz_predprey1	267	2	—	—	—	—	—	—
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9865	0.9827	0.9658	1.0750
fc3812567b5fdd64	197_cpu_act	800	21	5,006	pass	0.8164	0.9529	0.6632	1.1726
fe6d7ac3fcd3129e	1201_BNG_breastTumor	800	9	—	—	—	—	—	—

APPENDIX C

BEYONDARENA FAMILIES IN THE DATASET BUCKET

TABLE IX: All 142 BeyondArena families.

name	task	rows	cols	license	split
5g_energy_consumption	regression	92,629	22	MIT	grouped
acquire_valued_shoppers_challenge	binary	160,057	112	not cleared	temporal
airfoil_self_noise	regression	1,503	6	CC-BY-4.0	iid
allstate_claims_severity	regression	188,317	131	not cleared	iid
amazon_employee_access	binary	32,769	10	public-domain	iid
amex_non_iid	binary	1,249,605	191	not cleared	grouped
anes_voting_2026	binary	48,587	319	not cleared	temporal
aps_failure	binary	76,000	171	CC-BY-4.0	iid
asp_potassco_classification	multiclass	1,212	138	not cleared	grouped
audiology_diagnosis	multiclass	199	69	CC-BY-4.0	iid
bad_customer_detection	binary	1,723	14	public-domain	iid
bank_customer_churn	binary	10,000	11	public-domain	iid
bank_marketing	binary	45,211	14	CC-BY-4.0	iid
biogeographical_ancestry_prediction	multiclass	635	105	not cleared	iid
biomechanical_orthopaedic_prediction	multiclass	310	7	CC-BY-4.0	iid
bioresponse	binary	3,751	1777	public-domain	iid
blood_tests_drink_prediction	regression	345	6	CC-BY-4.0	iid
blood_transfusion	binary	748	5	CC-BY-4.0	iid
body_density_prediction	regression	252	14	not cleared	iid
california_house_prices_2020	regression	41,528	42	not cleared	temporal
cardiotocography	multiclass	2,126	24	CC-BY-4.0	grouped
churn	binary	5,000	20	public-domain	iid
cirrhosis_patient_survival_prediction	regression	161	18	not cleared	iid
climate_model_weather_forecasting	regression	1,250,000	101	not cleared	temporal
clock_protein_toxicity	binary	171	1118	CC-BY-4.0	iid
coffee_rating_prediction	regression	2,369	13	not cleared	temporal
coil_2000	binary	9,822	86	CC-BY-4.0	iid
concrete_compressive_strength	regression	1,030	9	CC-BY-4.0	iid
consumer_complaints	multiclass	1,226,140	13	us-gov	temporal
cooking_time	regression	1,250,000	197	not cleared	temporal
covertype	multiclass	512,625	15	CC-BY-4.0	grouped
credit_approval	binary	690	16	CC-BY-4.0	iid
credit_card_clients_default	binary	30,000	24	CC-BY-4.0	iid
credit_g	binary	1,000	21	CC-BY-4.0	iid
customer_satisfaction_in_airline	binary	129,880	22	public-domain	iid
delivery_eta	regression	1,250,000	226	not cleared	temporal
dementia_prediction	multiclass	370	10	not cleared	grouped
diabetes_130_us	binary	71,518	45	CC-BY-4.0	iid
diamonds	regression	53,940	10	MIT	iid
drug_induced_autoimmunity_prediction	binary	597	178	CC-BY-4.0	iid
early_learning_predictors	regression	18,874	745	CC-BY-4.0	grouped
early_stage_diabetes_risk_prediction	binary	251	17	CC-BY-4.0	iid
ecoli_proteins	multiclass	327	7	CC-BY-4.0	iid
ecommerce_shipping	binary	10,999	11	public-domain	iid
electric_motor_temperature_prediction	regression	1,296,316	111	not cleared	grouped
emscad	binary	17,460	18	CC0-1.0	iid
erythemato_squamous_disease	multiclass	366	35	CC-BY-4.0	iid
fiat_500	regression	1,538	8	CC0-1.0	iid
fitness_club	binary	1,500	7	public-domain	iid
food_delivery_time	regression	45,451	10	not cleared	iid
forensic_glass_identification	multiclass	214	10	CC-BY-4.0	iid
forest_fires	regression	517	13	CC-BY-4.0	iid
gallstone_disease	binary	319	39	CC-BY-4.0	iid
garments_worker_productivity	regression	1,197	16	CC-BY-4.0	temporal
ghanas_indigenous_intel	multiclass	10,928	11	not cleared	temporal
give_me_some_credit	binary	150,000	11	not cleared	iid
hazelnut_spread_contaminant_detection	binary	2,400	31	not cleared	iid
healthcare_insurance_expenses	regression	1,338	7	not cleared	iid
heart_disease_cleveland	binary	303	14	CC-BY-4.0	iid
heart_disease_hungary	binary	294	14	CC-BY-4.0	iid
heart_disease_va_long_beach	binary	200	14	CC-BY-4.0	iid
heart_failure_followup_survival	binary	299	13	CC-BY-4.0	iid
heloc	binary	10,459	24	not cleared	iid
hepatitis_c_prediction	multiclass	608	13	CC-BY-4.0	iid
hepatitis_survival_prediction	binary	155	20	CC-BY-4.0	iid
hiva_agnostic	binary	3,845	1519	public-domain	iid
home_credit_default_risk	binary	307,507	505	not cleared	iid

TABLE IX, continued

name	task	rows	cols	license	split
home_credit_default_stability	binary	1,224,927	712	not cleared	temporal
homeq_default_prediction	binary	5,708	13	not cleared	iid
homesite_quote_conversion	binary	260,753	296	not cleared	iid
horse_colic_survival	multiclass	344	21	CC-BY-4.0	iid
hotel_booking_demand	binary	81,418	32	CC-BY-4.0	temporal
houses	regression	19,675	9	not cleared	iid
hr_analytics	binary	19,158	13	CC0-1.0	iid
ieee_fraud_detection	binary	590,540	436	not cleared	temporal
immoscout_german_house_prices	regression	10,317	24	not cleared	iid
in_vehicle_coupon_recommendation	binary	12,684	25	CC-BY-4.0	iid
indian_liver_patient_dataset	binary	583	11	CC-BY-4.0	iid
iranian_churn	binary	2,850	14	CC-BY-4.0	iid
jml	binary	10,885	22	public-domain	iid
kdd_cup_09_appetency	binary	50,000	213	public-domain	iid
kick	binary	72,983	33	public-domain	temporal
kickstarter	binary	187,118	16	not cleared	temporal
labour_inspection_compliance	binary	63,634	377	CC0-1.0	iid
lending_club	binary	1,064,751	97	CC0-1.0	temporal
ljubljana_breast_cancer	binary	286	10	CC-BY-4.0	iid
ljubljana_primary_tumor	multiclass	302	18	CC-BY-4.0	iid
lung_cancer	multiclass	197	12601	not cleared	iid
lung_cancer_epithelial_genexp	binary	187	22216	public-domain	iid
maps_router_eta	regression	1,250,000	989	not cleared	temporal
marketing_campaign	binary	2,240	26	CC-BY-4.0	iid
maternal_health_risk	multiclass	1,014	7	CC-BY-4.0	iid
mercari_price_suggestion	regression	1,250,000	7	not cleared	iid
mercedes_benz_greener_manufacturing	regression	4,204	372	not cleared	temporal
miami_housing	regression	13,776	16	not cleared	iid
mic	multiclass	1,699	112	CC-BY-4.0	iid
mice_protein_trisomy_discriminant	multiclass	1,080	78	CC-BY-4.0	grouped
micro_mass	multiclass	571	1084	CC-BY-4.0	grouped
musk	binary	6,598	168	CC-BY-4.0	grouped
mutual_funds_india	regression	793	13	CC0-1.0	iid
naticusdroid_android_permissions_dataset	binary	7,491	86	CC-BY-4.0	iid
obesity_estimation	regression	498	15	CC-BY-4.0	iid
online_shoppers_purchasing_intention_dataset	binary	12,330	18	CC-BY-4.0	iid
otto_group_product_classification_challenge	multiclass	61,878	94	not cleared	iid
pancreatic_cancer_mouse_detection	binary	181	6773	not cleared	grouped
parkinsons_biomedical_voice_measurements	binary	195	25	CC-BY-4.0	grouped
physiochemical_protein	regression	45,730	10	CC-BY-4.0	iid
polish_companies_bankruptcy	binary	5,790	65	CC-BY-4.0	iid
porto_seguro	binary	595,206	38	not cleared	iid
predict_students_dropout_and_academic_success	multiclass	4,424	37	CC-BY-4.0	iid
prostate_cancer_detection	binary	322	15155	not cleared	iid
pva_revenue_prediction_kddcup98	binary	144,095	478	not cleared	iid
qsar_aquatic_toxicity	regression	546	9	CC-BY-4.0	iid
qsar_biodeg	binary	1,054	42	CC-BY-4.0	iid
qsar_fish_toxicity	regression	908	7	CC-BY-4.0	iid
qsar_tid_11	regression	5,741	1025	public-domain	iid
regensburg_pediatric_appendicitis	binary	763	52	not cleared	iid
rossmann_store_sales	regression	844,392	16	not cleared	temporal
santander_customer_satisfaction	binary	71,080	308	not cleared	iid
santander_customer_transaction_prediction	binary	200,000	601	not cleared	iid
santander_transaction_value	regression	4,447	541	not cleared	iid
sat11_hand_algo_runtime	regression	2,960	171	not cleared	grouped
sberbank_housing_market_forecasting	regression	27,195	387	not cleared	temporal
sdss_17	multiclass	78,053	12	public-domain	iid
seismic_bumps	binary	2,584	16	CC-BY-4.0	iid
sepsis_prediction	binary	1,228,686	44	not cleared	grouped
sepsis_survival_minimal_clinical_records	binary	110,204	4	CC-BY-4.0	iid
sf_permit_time	regression	116,954	38	not cleared	temporal
south_africa_coronary_heart_disease	binary	462	10	not cleared	iid
splice	multiclass	3,190	61	CC-BY-4.0	iid
student_portuguese_performance	regression	649	31	CC-BY-4.0	iid
superconductivity	regression	21,263	82	CC-BY-4.0	iid
taiwanese_bankruptcy_prediction	binary	6,819	93	CC-BY-4.0	iid
telemonitoring_parkinsons_biomedical_voice_measurements	regression	502	21	CC-BY-4.0	grouped
thyroid_discordant	binary	3,711	27	CC-BY-4.0	iid
tour_travels_churn	binary	954	7	CC0-1.0	iid
video_game_fps_prediction	regression	12,288	40	CC-BY-4.0	grouped
video_transcoding_time_prediction	regression	68,784	20	CC-BY-4.0	grouped
website_phishing	multiclass	1,353	10	CC-BY-4.0	iid
wids_diabetes_mellitus	binary	127,358	182	not cleared	iid
wine_quality	regression	6,497	13	CC-BY-4.0	iid
wine_world_cost	regression	1,279	15	CC0-1.0	iid

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